

E070

Front End Development

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A library director, who has run the branch for 11 years, tells the web team: “Members should be able to renew books online easily.” The library has 30,000 members. What should the team write before building?

- A. A user story with testable acceptance criteria, such as renewing a book from the account page in three clicks or fewer.
- B. A requirement that says the site must be user friendly.
- C. A plan to scale servers for 30,000 members.
- D. A database schema for renewals.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

A café chain’s site, designed on a 27-inch monitor, uses a fixed 1200 px container. On phones it scrolls sideways. The chain has 40 locations. What approach fixes this best?

- A. Make a separate phone site for each location.
- B. Raise the container to 1400 px.
- C. Tell users to rotate their phones.
- D. Use fluid widths (for example max-width: 100%) with mobile-first styles and min-width media queries.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3.

A news site built by a team in Denver needs a page with a header, sidebar, main area, and footer, where columns and rows must line up. The team uses a design system with 14 color tokens. Which CSS tool fits best?

- A. Floats
- B. Nested Flexbox rows only
- C. CSS Grid
- D. position: absolute on every section

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4. [TRANSFER — reversed stem / different variable]

A toolbar needs five buttons spaced evenly in one row, with the row wrapping on small screens. Which tool is the simplest fit?

- A.** A fixed pixel margin between each button
- B.** An HTML table
- C.** CSS Grid with named areas
- D.** Flexbox with flex-wrap

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A clothing store's menu uses div elements with click handlers as buttons. The store just rebranded with a new logo. Keyboard users cannot tab to the items, and screen readers do not announce them. What is the best fix?

- A.** Replace the divs with native button elements.
- B.** Add ARIA to every element on the page.
- C.** Make the new logo larger.
- D.** Add role='button' to each div and nothing else.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A signed-in user of an HR portal, hosted in a data center in Ohio, opens the payroll admin page and the API returns 403. The user's login token is valid. What does 403 mean here?

- A.** The user is not signed in.
- B.** The server knows who the user is, but the user is not allowed to access this resource.
- C.** The server crashed.
- D.** The page does not exist.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A front end must add a new comment to a post through a REST API. Which method and success code fit best?

- A.** GET, returning 200 OK
- B.** DELETE, returning 204
- C.** POST, returning 201 Created
- D.** POST, returning 404

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A weather dashboard calls `fetch(url).then(r => r.json()).then(show).catch(showError)`. The API is run by a partner in Canada. When the API returns 500, no error message appears and the page shows garbage. Why?

- A. Replace `.json()` with `.text()`.
- B. `fetch` only rejects on network failure; check `response.ok` and throw when it is false.
- C. The partner's location blocks error codes.
- D. `.catch` always runs on any error, so the API must be returning 200.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In which case does a `fetch` promise reject and trigger `.catch`?

- A. The server returns 404 Not Found.
- B. The response body is empty.
- C. The device is offline and the request cannot reach any server.
- D. The server returns 500.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student app at `app.example.edu` calls an API at `api.partner.com`. The request works from a command-line tool but the browser blocks it with a CORS error. The team has four developers and a sprint ending Friday. Where must the fix happen?

- A. On the API server, which must return an `Access-Control-Allow-Origin` header allowing the app's origin (or use a same-origin proxy).
- B. Add developers so the sprint finishes on time.
- C. In the browser, by telling users to turn off security.
- D. In the front end, by adding an `Access-Control-Allow-Origin` request header.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which request needs no CORS permission?

- A. `https://shop.com` calling `https://api.shop.com`
- B. `https://shop.com` calling `http://shop.com`
- C. `https://shop.com` calling `https://shop.com:8443`
- D. A page at `https://shop.com` calling `https://shop.com/api/items`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A travel site adds a paid mapping API. A developer places the private API key in the front-end JavaScript bundle and minifies it. The site gets 5,000 visitors a day. What is the problem and fix?

- A. Put the key in an HTML comment instead.
- B. The key is exposed to anyone; call the API through a back-end service that holds the key.
- C. Limit the site to fewer visitors.
- D. Minifying hides the key well enough.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A library director, ~~who has run the branch for 11 years,~~ tells the web team: “Members should be able to renew books online easily.” ~~The library has 30,000 members.~~ What should the team write before building?

Target word: turn a request into buildable work **Decides:** “Easily” is not testable.

A	Correct	Clear who/what/why plus a pass/fail check.
B	Dropped element	Still vague and untestable.
C	Camouflage	Member count does not define the feature.
D	Adjacent category	Design work, done after the requirement is defined.

2. Answer D Concept 2

A café chain’s site, ~~designed on a 27-inch monitor,~~ uses a fixed 1200 px container. On phones it scrolls sideways. ~~The chain has 40 locations.~~ What approach fixes this best?

Target word: responsive strategy **Decides:** Fixed pixel width does not adapt.

A	Camouflage	Location count does not require separate sites.
B	Inverted operation	Makes the overflow worse.
C	Dropped element	Leaves the layout broken.
D	Correct	The layout adapts from small screens up.

3. Answer C Concept 3

A news site ~~built by a team in Denver~~ needs a page with a header, sidebar, main area, and footer, where columns and rows must line up. ~~The team uses a design system with 14 color tokens.~~ Which CSS tool fits best?

Target word: two-dimensional layout choice **Decides:** Rows and columns must align together.

A	Adjacent category	An older technique not meant for full page layout.
B	Confusion set	Flexbox handles one axis; aligning both takes workarounds.
C	Correct	Controls rows and columns at once.
D	Absolute language	Breaks normal flow and responsiveness.

4. Answer D Concept 3 · transfer

A toolbar needs five buttons spaced evenly in one row, with the row wrapping on small screens. Which tool is the simplest fit?

Target word: one-dimensional layout **Decides:** A single row of items.

A	Dropped element	No even spacing or wrapping.
B	Adjacent category	Tables are for tabular data.
C	Inverted operation	A two-dimensional tool for a one-row job.
D	Correct	Built for spacing items along one axis.

5. Answer A Concept 4

A clothing store's menu uses div elements with click handlers as buttons. ~~The store just rebranded with a new logo.~~ Keyboard users cannot tab to the items, and screen readers do not announce them. What is the best fix?

Target word: accessible control choice **Decides:** Native buttons give focus and roles automatically.

A	Correct	Buttons are focusable, keyboard-operable, and announced.
B	Absolute language	Extra ARIA can make things worse; native first.
C	Camouflage	The rebrand does not affect the menu.
D	Dropped element	Still not focusable or keyboard-operable.

6. Answer B Concept 5

A signed-in user of an HR portal, ~~hosted in a data center in Ohio,~~ opens the payroll admin page and the API returns 403. The user's login token is valid. What does 403 mean here?

Target word: meaning of 403 **Decides:** The user is identified but lacks permission.

A	Confusion set	That is 401 Unauthorized.
B	Correct	403 Forbidden is an authorization failure.
C	Inverted operation	Server faults are 5xx; 4xx points at the request.
D	Adjacent category	That is 404.

7. Answer C Concept 5 · transfer

A front end must add a new comment to a post through a REST API. Which method and success code fit best?

Target word: method and status for create **Decides:** Creating a resource.

A	Inverted operation	GET reads and must not create.
B	Confusion set	Removes a resource.
C	Correct	POST creates; 201 confirms creation.
D	Adjacent category	404 means not found, not success.

8. Answer B Concept 6

A weather dashboard calls `fetch(url).then(r => r.json()).then(show).catch(showError)`. ~~The API is run by a partner in Canada.~~ When the API returns 500, no error message appears and the page shows garbage. Why?

Target word: why `.catch` did not run **Decides:** `fetch` resolves on HTTP errors.

A	Adjacent category	Changes parsing, not error detection.
B	Correct	A 500 still resolves, so <code>.catch</code> is skipped.
C	Camouflage	Geography does not change <code>fetch</code> behavior.
D	Absolute language	The premise is wrong for <code>fetch</code> .

9. Answer C Concept 6 · transfer

In which case does a `fetch` promise reject and trigger `.catch`?

Target word: when `fetch` rejects **Decides:** Only network-level failure rejects.

A	Inverted operation	Resolves with <code>ok = false</code> .
B	Adjacent category	<code>fetch</code> still resolves; parsing may fail later.
C	Correct	A network failure rejects.
D	Confusion set	Also resolves with <code>ok = false</code> .

10. Answer A Concept 7

A student app at app.example.edu calls an API at api.partner.com. The request works from a command-line tool but the browser blocks it with a CORS error. ~~The team has four developers and a sprint ending Friday.~~ Where must the fix happen?

Target word: where CORS is fixed **Decides:** The browser needs permission from the API server's response headers.

A	Correct	CORS permission comes from the server.
B	Camouflage	Team size does not fix headers.
C	Absolute language	Unsafe and not a real fix.
D	Inverted operation	That header must be in the response, not the request.

11. Answer D Concept 7 · transfer

Which request needs no CORS permission?

Target word: when CORS does not apply **Decides:** Same scheme, host, and port = same origin.

A	Confusion set	A different subdomain is a different origin.
B	Dropped element	Different scheme, different origin.
C	Adjacent category	Different port, different origin.
D	Correct	Same origin; CORS does not apply.

12. Answer B Concept 8

A travel site adds a paid mapping API. A developer places the private API key in the front-end JavaScript bundle and minifies it. ~~The site gets 5,000 visitors a day.~~ What is the problem and fix?

Target word: secret handling **Decides:** Everything in the bundle is readable by users.

A	Adjacent category	Still shipped to the browser.
B	Correct	Secrets stay on the server.
C	Camouflage	Traffic does not change exposure.
D	Confusion set	Minified code is still readable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	User stories with acceptance criteria	Capture a requirement as who needs what and why, plus testable acceptance criteria that define when it is done.	Writing vague goals (“make it user friendly”) that no one can test.	A stakeholder request that must be turned into buildable work.
2	Responsive, mobile-first layout	Write base styles for small screens, then add min-width media queries and relative units as the screen grows.	Building a fixed-width desktop page and shrinking it later.	A layout that breaks or scrolls sideways on phones.
3	Flexbox vs. Grid	Flexbox lays out items along one axis (a row or a column); Grid controls rows and columns together.	Forcing a two-dimensional page layout with nested flex rows.	Whether both rows and columns must line up.
4	Semantic HTML first, ARIA second	Use native elements (button, label, nav, alt text) for built-in accessibility; add ARIA only when no native element fits.	Making a clickable div and patching it with ARIA.	A custom control that keyboard or screen-reader users cannot use.
5	HTTP methods and status codes	GET reads, POST creates, PUT/PATCH update, DELETE removes; 2xx success, 4xx client problem (401 not authenticated, 403 not allowed, 404 not found), 5xx server problem.	Mixing up 401 and 403, or blaming the server for a 4xx.	A status code in a failing API call.
6	fetch error handling	fetch rejects only on network failure; an HTTP error such as 404 or 500 still resolves, so check response.ok before using the body.	Assuming .catch will run on every failed request.	An error page parsed as data, or no error shown for a 500.
7	CORS	The browser blocks cross-origin responses unless the API server sends the right Access-Control-Allow-Origin header; the fix is on the server (or a same-origin proxy).	Trying to fix CORS with front-end code or request headers.	Works in a tool like curl, blocked in the browser.
8	Keep secrets off the client	Anything shipped to the browser is visible to users, so private API keys belong on a server that calls the API for the front end.	Hiding a key in a minified bundle or an environment file that gets bundled.	A paid or private API key used from front-end code.

Course scope (public description): This course develops learners’ ability to analyze user and stakeholder requirements and design effective web-based solutions using existing frameworks, assets, and web content. Learners apply computing principles and contemporary tools to implement user-centered designs that emphasize usability, accessibility, and functional page layout from multiple user perspectives. The course further strengthens learners’ skills in integrating application programming interfaces (APIs) and web services to add data-driven capabilities to web applications. Through applied design and implementation activities, learners demonstrate the ability to solve computing problems, evaluate design tradeoffs, and develop web solutions that meet specified functional and user requirements, consistent with professional computing practice.